

Part A: Square and Cube Roots*Evaluate each root.*

1. $\sqrt{49}$

6. $\sqrt{4}$

2. $\sqrt{9}$

7. $\sqrt[3]{8}$

3. $\sqrt{16}$

8. $\sqrt[3]{27}$

4. $\sqrt{25}$

9. $\sqrt[3]{64}$

5. $\sqrt{36}$

Part B: Simplifying Square and Cube Roots*Simplify each radical completely.*

1. $\sqrt{54}$

7. $\sqrt[3]{72}$

2. $\sqrt{52}$

8. $\sqrt[3]{96}$

3. $\sqrt{104}$

9. $\sqrt[3]{112}$

4. $\sqrt{96}$

10. $\sqrt[3]{104}$

5. $\sqrt{56}$

11. $\sqrt[3]{56}$

6. $\sqrt{68}$

12. $\sqrt[3]{48}$

Part C: Negative Number Operations*Evaluate each expression completely.*

1. $-16 - (-9)$

3. $-13 + 17$

2. $9 - 2$

4. $-16 - 4$

5. $-26 - (-37)$

15. $-104 \div 13$

6. $25 - 9 - (-32)$

16. $-39 \div 13$

7. $38 - 10 - (-25)$

17. $8 + (-4) \cdot 9$

8. $-36 - 14 - (-20)$

18. $-20 \div (-4) - (-2)$

9. $-4 \cdot (-8)$

19. $-7 + 4 \cdot (-2)$

10. $12 \cdot (-4)$

20. $-50 \div (-10) - 3$

11. $-12 \cdot (-12)$

21. $-36 + 9 \cdot (-12)$

12. $-70 \div 10$

22. $-14 + 6 \cdot (-12)$

13. $-72 \div 8$

23. $54 \div (-9) - (-36)$

14. $-3 \cdot 7$

24. $-70 \div 5 - (-4)$

Part D: Order of Operations

Evaluate each expression completely. Show each major step.

1. $(-7)^3 - 8 \cdot 5$

5. $(-5)^2 - (-15) \cdot 4$

2. $(-8)^3 - 8 \cdot 7$

6. $(-3)^2 - (-16) \cdot 11$

3. $(4 + (-2))^2 \div (-1)$

7. $19 - 6 \cdot (3 + 19)$

4. $9^3 - 16 \cdot 9$

8. $-8 - 8 \cdot ((-6) + (-8))$

9. $(-4)^2 - (-10) \cdot 13$

13. $(5 + 0)^2 \div (-1)$

10. $(5 + 18)^2 \div 1$

14. $3 - 3 \cdot (2 + 3)$

11. $11 - 7 \cdot ((-4) + 11)$

15. $-16 - 9 \cdot (5 + (-16))$

12. $(-8 + 9)^2 \div (-1)$

16. $29 - 8 \cdot ((-9) + 29)$

Part E: Solving Linear Equations

Solve each equation for x . Show your steps.

1. $7x + 12 = -37$

7. $4(9x - 10) + 3 = -145$

2. $11x - 3 = -124$

8. $5(4x + 10) - 7 = -177$

3. $3x + 2 = -34$

9. $8(6x + 2) - 8 = -280$

4. $6x + 5 = 59$

10. $8(6x + 8) + 3 = 163$

5. $7x + 11 = \frac{79}{4}$

11. $-7x - 12 = -8x - 17$

6. $4(7x - 6) + 8 = 208$

12. $5x - 2 = -20x + 273$

13. $-16x + 12 = -15x + \frac{55}{3}$

17. $-\frac{2}{3}x + 16 = \frac{58}{3}$

14. $-20x + 6 = -7x - 150$

18. $\frac{x}{3} - 15 = -\frac{61}{3}$

15. $-17x + 1 = x - 35$

19. $\frac{x+2}{4} + 12 = \frac{41}{4}$

16. $\frac{x+16}{2} + 10 = \frac{49}{2}$

20. $\frac{x}{3} - 17 = -\frac{82}{5}$

Part F: Numerical Greatest Common Factor

Find the greatest common factor.

1. GCF (42, 6)

6. GCF (42, 36)

2. GCF (24, 21)

7. GCF (15, 24)

3. GCF (24, 30)

8. GCF (108, 120)

4. GCF (5, 35)

9. GCF (108, 60)

5. GCF (40, 5)

10. GCF (54, 24)

Part G: Greatest Common Factor of Monomials

Find the greatest common factor of each pair of monomials.

1. GCF ($9x^1, 15x^2$)

3. GCF ($10x^3, 16x^2$)

2. GCF ($16x^3, 6x^2$)

4. GCF ($12x^3, 18x^2$)

5. GCF $(14x^2, 4x^1)$

9. GCF $(28x^5, 77x^1)$

6. GCF $(48x^1, 18x^3)$

10. GCF $(80x^5, 110x^3)$

7. GCF $(12x^1, 32x^3)$

11. GCF $(30x^2, 21x^3)$

8. GCF $(56x^1, 16x^2)$

12. GCF $(45x^4, 55x^1)$

Part H: Combining Linear Like Terms

Combine like terms. Operate on coefficients and keep the variable part unchanged.

1. $7x + 11 + 6x + 12$

5. $3x + 12 + 5x + 4$

2. $7x + 12 - 7x - 3$

6. $5x + 7 + 10x + 8$

3. $11x + 7 + 2x + 7$

7. $10x + 5 + 8x - 11$

4. $12x + 7 + 10x + 9$

8. $12x - 10 + 8x + 9$

Part I: Combining Polynomial Like Terms

Combine like terms. Do not add or multiply the exponents.

1. $4x^2 + 2x - 4x + 12x^2$

5. $-6x^2 + 5x + 3x^2 + 9x$

2. $11x^2 + 11x + 5x^2 + 11x$

6. $3x + 10x + 8x^2 + 11x^2$

3. $12x + 6x^2 + 4x - 4x^2$

7. $4x^2 + 3x + 8x^2 + 5x$

4. $9x + x^2 + 6x^2 - 4x$

8. $12x^2 + 10x^2 - 5x - 4x$

9. $-12x^2 + 15x^3 - 3x^3 + 8x + 11x^2 + 5x$

13. $3x^2 + 10x - 10x^2 + 13x^3 + 7x + 9x^3$

10. $11x - 12x^3 - 3x^2 - 20x^3 - 3x^2 - 15x$

14. $16x^3 + 7x + 14x^2 + 12x^2 + 18x^3 + 10x$

11. $-17x^2 + 8x^3 + 6x^3 + 4x^2 - 20x - 8x$

15. $3x^2 + 4x^2 + 9x^3 + 2x + 8x^3 + 12x$

12. $3x^2 + 20x + 15x^3 + 3x + 5x^2 - 8x^3$

16. $3x - 2x^3 + 20x^3 + 9x^2 + 16x^2 + 2x$

Part J: Exponent Properties

Simplify. Exponent rules apply to multiplication, division, and powers—not addition.

1. $2x^7 \cdot 4x^8$

9. $(4x^9)^3$

2. $4x^3 \cdot 5x^4$

10. $(5x^6)^3$

3. $4x^9 \cdot 7x^3$

11. $(6x^9)^2$

4. $8x^5 \cdot 9x^2$

12. $(3x^3)^2$

5. $\frac{5x^{16}}{x^8}$

13. $\frac{10x^9y^{13}}{5x^7y^7}$

6. $\frac{6x^{10}}{x^6}$

14. $\left(\frac{5x^8}{5x^5}\right)^3$

7. $\frac{4x^{11}}{x^8}$

15. $\left(\frac{6x^8}{3x^4}\right)^2$

8. $\frac{2x^8}{x^4}$

16. $(11x^5y^7)^2 \cdot 9x^6y^1$

17. $(6x^2y^3)^3 \cdot 6x^2y^4$

18. $\frac{9x^7y^{12}}{x^2y^2}$

Part K: Factoring Linear Expressions

Factor out the greatest common factor. Check by distributing.

1. $20x + 12$

5. $24x + 4$

2. $-3x - 12$

6. $25x - 45$

3. $-40x + 48$

7. $4x - 36$

4. $56x - 35$

8. $7x + 42$

Answer Key

Part A: Square and Cube Roots

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|------|------|------|
| 1. 7 | 4. 5 | 7. 2 |
| 2. 3 | 5. 6 | 8. 3 |
| 3. 4 | 6. 2 | 9. 4 |

Part B: Simplifying Square and Cube Roots

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|-----------------|--------------------|---------------------|
| 1. $3\sqrt{6}$ | 5. $2\sqrt{14}$ | 9. $2\sqrt[3]{14}$ |
| 2. $2\sqrt{13}$ | 6. $2\sqrt{17}$ | 10. $2\sqrt[3]{13}$ |
| 3. $2\sqrt{26}$ | 7. $2\sqrt[3]{9}$ | 11. $2\sqrt[3]{7}$ |
| 4. $4\sqrt{6}$ | 8. $2\sqrt[3]{12}$ | 12. $2\sqrt[3]{6}$ |

Part C: Negative Number Operations

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|--------|---------|----------|
| 1. -7 | 9. 32 | 17. -28 |
| 2. 7 | 10. -48 | 18. 7 |
| 3. 4 | 11. 144 | 19. -15 |
| 4. -20 | 12. -7 | 20. 2 |
| 5. 11 | 13. -9 | 21. -144 |
| 6. 48 | 14. -21 | 22. -86 |
| 7. 53 | 15. -8 | 23. 30 |
| 8. -30 | 16. -3 | 24. -10 |

Part D: Order of Operations

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|---------|---------|----------|
| 1. -383 | 7. -113 | 13. -25 |
| 2. -568 | 8. 104 | 14. -12 |
| 3. -4 | 9. 146 | 15. 83 |
| 4. 585 | 10. 529 | 16. -131 |
| 5. 85 | 11. -38 | |
| 6. 185 | 12. -1 | |

Part E: Solving Linear Equations

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|----------------------|-------------------------|-----------------------|
| 1. $x = -7$ | 8. $x = -11$ | 15. $x = 2$ |
| 2. $x = -11$ | 9. $x = -6$ | 16. $x = 13$ |
| 3. $x = -12$ | 10. $x = 2$ | 17. $x = -5$ |
| 4. $x = 9$ | 11. $x = -5$ | 18. $x = -16$ |
| 5. $x = \frac{5}{4}$ | 12. $x = 11$ | 19. $x = -9$ |
| 6. $x = 8$ | 13. $x = -\frac{19}{3}$ | 20. $x = \frac{9}{5}$ |
| 7. $x = -3$ | 14. $x = 12$ | |

Part F: Numerical Greatest Common Factor

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|------|-------|-------|
| 1. 6 | 5. 5 | 9. 12 |
| 2. 3 | 6. 6 | 10. 6 |
| 3. 6 | 7. 3 | |
| 4. 5 | 8. 12 | |

Part G: Greatest Common Factor of Monomials

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|-----------|---------|-------------|
| 1. $3x$ | 5. $2x$ | 9. $7x$ |
| 2. $2x^2$ | 6. $6x$ | 10. $10x^3$ |
| 3. $2x^2$ | 7. $4x$ | 11. $3x^2$ |
| 4. $6x^2$ | 8. $8x$ | 12. $5x$ |

Part H: Combining Linear Like Terms

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|---------------|---------------|--------------|
| 1. $13x + 23$ | 4. $22x + 16$ | 7. $18x - 6$ |
| 2. 9 | 5. $8x + 16$ | 8. $20x - 1$ |
| 3. $13x + 14$ | 6. $15x + 15$ | |

Part I: Combining Polynomial Like Terms

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|------------------|---------------------------|---------------------------|
| 1. $16x^2 - 2x$ | 7. $12x^2 + 8x$ | 13. $22x^3 - 7x^2 + 17x$ |
| 2. $16x^2 + 22x$ | 8. $22x^2 - 9x$ | 14. $34x^3 + 26x^2 + 17x$ |
| 3. $2x^2 + 16x$ | 9. $12x^3 - x^2 + 13x$ | 15. $17x^3 + 7x^2 + 14x$ |
| 4. $7x^2 + 5x$ | 10. $-32x^3 - 6x^2 - 4x$ | 16. $18x^3 + 25x^2 + 5x$ |
| 5. $-3x^2 + 14x$ | 11. $14x^3 - 13x^2 - 28x$ | |
| 6. $19x^2 + 13x$ | 12. $7x^3 + 8x^2 + 23x$ | |

Part J: Exponent Properties

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|---------------|-----------------|------------------------|
| 1. $8x^{15}$ | 7. $4x^3$ | 13. $2x^2y^6$ |
| 2. $20x^7$ | 8. $2x^4$ | 14. x^9 |
| 3. $28x^{12}$ | 9. $64x^{27}$ | 15. $4x^8$ |
| 4. $72x^7$ | 10. $125x^{18}$ | 16. $1089x^{16}y^{15}$ |
| 5. $5x^8$ | 11. $36x^{18}$ | 17. $1296x^8y^{13}$ |
| 6. $6x^4$ | 12. $9x^6$ | 18. $9x^5y^{10}$ |

Part K: Factoring Linear Expressions

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|-----------------|----------------|---------------|
| 1. $4(5x + 3)$ | 4. $7(8x - 5)$ | 7. $4(x - 9)$ |
| 2. $3(-x - 4)$ | 5. $4(6x + 1)$ | 8. $7(x + 6)$ |
| 3. $8(-5x + 6)$ | 6. $5(5x - 9)$ | |